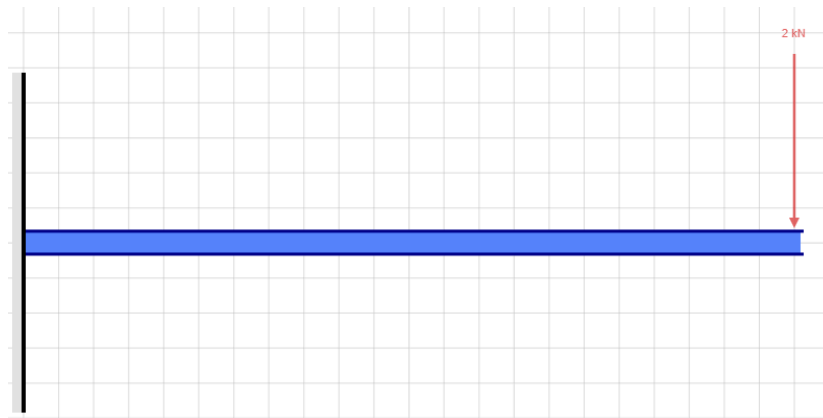


Part I.

1. The Free-Body Diagram of a rigid body represents it as having been isolated from its surroundings, replacing physical supports with forces and couple moments.
2. *True or False.* An ideal truss does not have three-force members. False.
3. For 3D systems, the internal moment has two components: torsion and bending whose axes are tangent and normal to the cross section, respectively.
4. A body is said to be Partially constrained if it experiences a fewer number of reactive forces than there are applicable equations of static equilibrium.
5. *True or False.* A beam is subject to a load uniformly distributed throughout its length. The beam is supported at its left end by a pin, at its right end by a rocker, and at its midpoint by a roller. The beam is statically determinate. False
6. For a simply supported thirteen-member plane truss to be considered simple, it must have 8 joints.
7. *True or False.* A frictionless ball-and-socket joint resists translations and rotations. False.
8. A body is considered improperly constrained if all the reactive forces intersect at a common point or pass through a common axis, or if all reactive forces are parallel.
9. *True or False.* A stable body requires that the lines of action of the reactive forces do not intersect a common axis and are not parallel to one another. True.
10. For internal loadings of 2D systems, the normal force and the shear force are normal and tangent to the cross section, respectively.

Scenario 1:

A 1.1-m long cantilever beam is subject to a 2-kN weight at its free end.

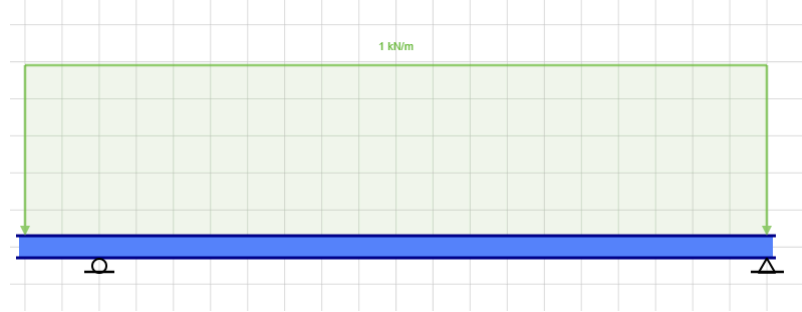


11. True or False. This scenario is statically determinate. True
12. The support reactions have magnitudes of 2 kN and 2.2 kN-m.
13. The maximum bending moment developed is 2.2 kN-m in magnitude and occurs 1.1 m from the free end.
14. The maximum shear force developed is 2 kN-m in magnitude and occurs 0 m from the free end.
15. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree 0
16. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree 1

Scenario 2.

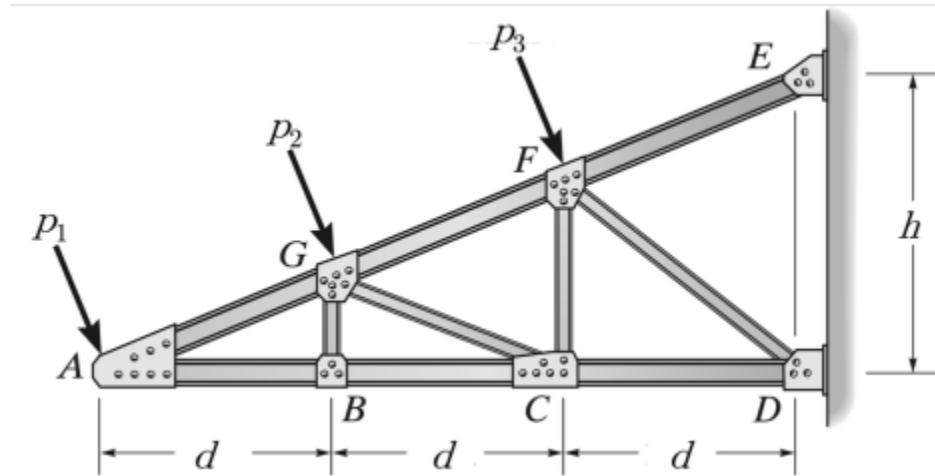
Group “Audentes! Fortuna Iuvat!

A 1-m long beam carries a load uniformly distributed throughout its length. Assume a roller support 0.1m from the left end and a pin support at the right end. The load has an intensity of 1 kN/m.

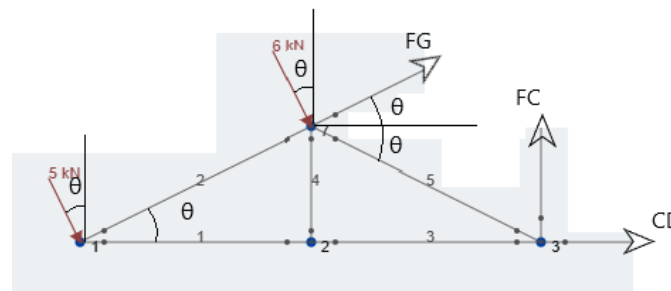


17. The roller support develops a force of 0.556 kN.
18. The pin support develops a force of 0.444 kN.
19. The bending moment developed at the right end is 0 kN-m.
20. The shear force developed just to the left of the pin is -0.444 kN,
21. The shear force developed just to the right of the pin is 0 kN,
22. True or False. There is zero shear force developed at a point 0.9 from the right end.
23. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree 1
24. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree 2

Problem 1. Consider the truss subject to idealized loads as shown above. Assume that the members have negligible weights and that the joints behave as frictionless pins. The magnitudes of P_1 , P_2 , and P_3 are 5 kN, 6 kN, and 7 kN, respectively. Set the dimensional parameters d and h to 2 ft and 3 ft, respectively.



- A. (15 points) Determine the forces developed in members FG, CF and CD. Use the method of sections, making sure to indicate the section cut and show the corresponding free-body diagram.



Angle of the larger triangle
 $\theta = \tan^{-1}(3/6) = 26.56505118^\circ$
 Length of opposite after cut:
 $2 \tan \theta = x, x = 1 \text{ m}$

To find CD and FC, it would be beneficial to calculate the summation of external forces or their components:

$$F_x = 5 \sin(26.56) + 6 \sin(26.56) + 14.5 \cos(26.56) + CD = 0$$

$$F_y = -5 \cos(26.56) - 6 \cos(26.56) + 14.5 \sin(26.56) + FC = 0$$

To minimize equations, do moment calculation at joint C:

$$M = -FG \cos(26.56)(1) - FG \sin(26.56)(2) - 6 \sin(26.56)(10 + 6 \cos(26.56)(2) + 5 \cos(26.56)(4) = 0$$

Solving yields:

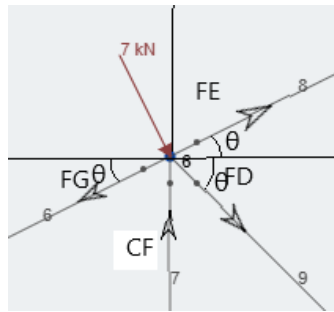
$$FG = 14.5 \text{ kN (Tension)}$$

Solving CD and FC yields:

$$CD = 17.889 \text{ kN (Compression)}$$

$$FC = 3.354 \text{ kN (Tension)}$$

- B. (10 points) Determine the forces developed in members DF and EF. Use the method of joints, making sure to show free-body diagrams at the joints considered. Hint: Use the computed forces in members FG, CF and CD.



To solve for FE and FD, do a method of Joints at Joint F where CF and FG values, as well as their "sense" are known.

$$F_x = FE \cos(26.56) + FD \cos(26.56) - 14.5 \cos(26.56) + 7 \sin(26.56) = 0 \text{ eq. 1}$$

$$F_y = FE \sin(26.56) - FD \sin(26.56) - 14.5 \sin(26.56) - 7 \cos(26.56) + 3.354 = 0 \text{ eq. 2}$$

Solving the 2 equations yields:

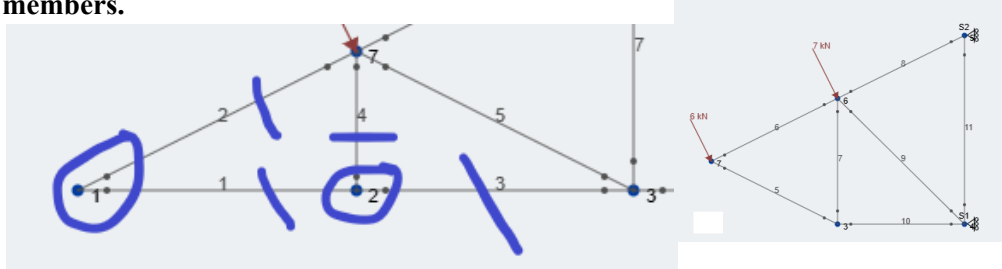
$$FE = 16 \text{ kN (Tension)}$$

$$FD = 5 \text{ kN (Compression)}$$

- C. (5 points) If the truss is to be idealized as simply supported by a pin and a roller, at which joints must the pin and the roller be placed to maintain stability of the loaded truss? Justify your answer.

- **The pin must be put at joint A and the roller at joint D, joint A gives a horizontal resistance to the loadings and its location primes it to resist the point loads particularly P1 which arguable produces the biggest forces in the x-direction due to its proximity, while the roller at joint E assures support with the vertical reactions making the truss stable.**

- D. (4 points) A contingency is considered wherein joint A is effectively unloaded. Identify zero-force members, if there be any, for this case. **AB, BG, AG and BC are zero force members.**



- E. (6 points) Determine the forces developed in the supports at E and at D. Hint: Use the computed forces in members CD, DF and EF.

Hence:

$$E_x = 14.311 \text{ kN, to the right}$$

$$D_x = 22.361 \text{ kN, to the left}$$

$$D_y = 9.391 \text{ kN, upward}$$

For a simple truss, there must be a pin and roller support so to maintain stability, assuming D has a pin and E has a roller (by virtue of location relative to the point loadings).

Problem explicitly stated to use Method of Joints:

Analyzing Joint E:

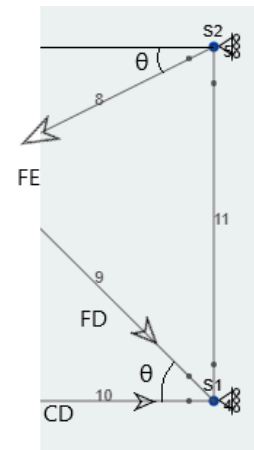
$$F_x = -E_x - 16 \cos(26.56) = 0, E_x = -14.31083506 \text{ kN}$$

$$F_y = -16 \sin(26.56) - DE = 0, DE = 7.155417528 \text{ kN (C)}$$

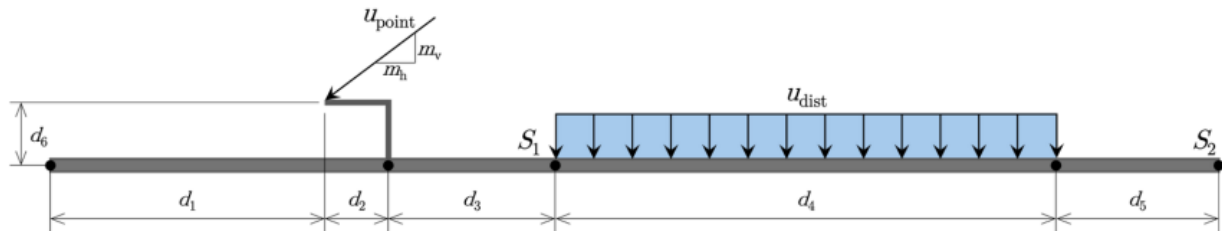
Analyzing Joint D:

$$F_x = -D_x + 17.889 + 5 \cos(26.56) = 0, D_x = 22.36113595 \text{ kN}$$

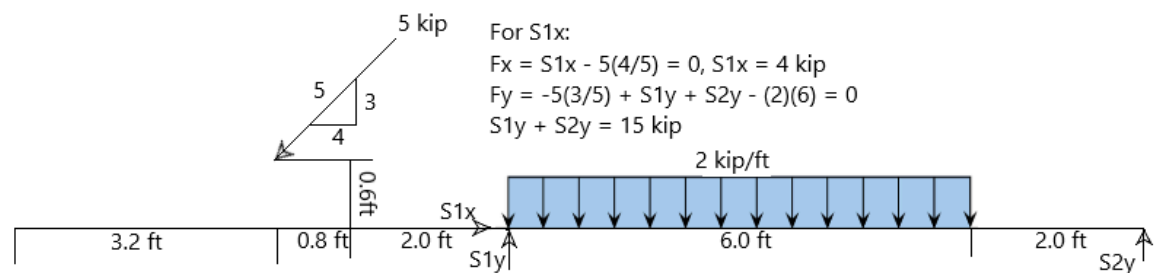
$$F_y = D_y - 5 \sin(26.56) - 7.155417528 = 0, D_y = 9.391485505 \text{ kN}$$



Problem 2. A custom beam design places supports at S1 and at S2, in anticipation of a concentrated load of u_{point} kips and a uniformly distributed load of intensity u_{dist} kip/ft. Your main task is to analyze how the internal loadings induced vary as a function of x , where r is the distance from the left end of the beam. Consider the case when U_{point} is 5 kips and U_{dist} is 2 kip. Set the respective values of d_1 , d_2 , d_3 , d_4 , d_5 and d_6 to 3.2, 0.8, 2.0, 6.0, 2.0, and 0.6 ft, and those of m_v and m_h to 3 and 4. For static determinacy, assume a pin at S1 and a roller at S2.



A. (6 points) Determine the support reactions.



Let's do a moment calculation at S1:

$$M = 0 = 5(4/5)(0.6) + 5(3/5)(2.8) + S2y(8) - (6)(2)(6/2)$$

$$S2y = 3.15 \text{ kip}$$

$$S1y = 11.85 \text{ kip}$$

$$S2x = 4 \text{ kip}$$

B. (1 point) Determine the normal force at $d_1 + d_2$, from the left end. **4 kip**

C. (2 points) Determine the shear force at $d_1 + d_2$ from the left end. **-3 kip**

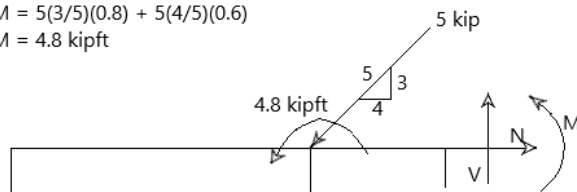
D. (3 points) Determine the bending moment at $d_1 + d_2$ from the left end. **-4.8 kipft**

For convenience, transfer Upoint to the point located at $d_1 + d_2$

To do so, calculate the moment Upoint creates at that point:

$$M = 5(3/5)(0.8) + 5(4/5)(0.6)$$

$$M = 4.8 \text{ kipft}$$



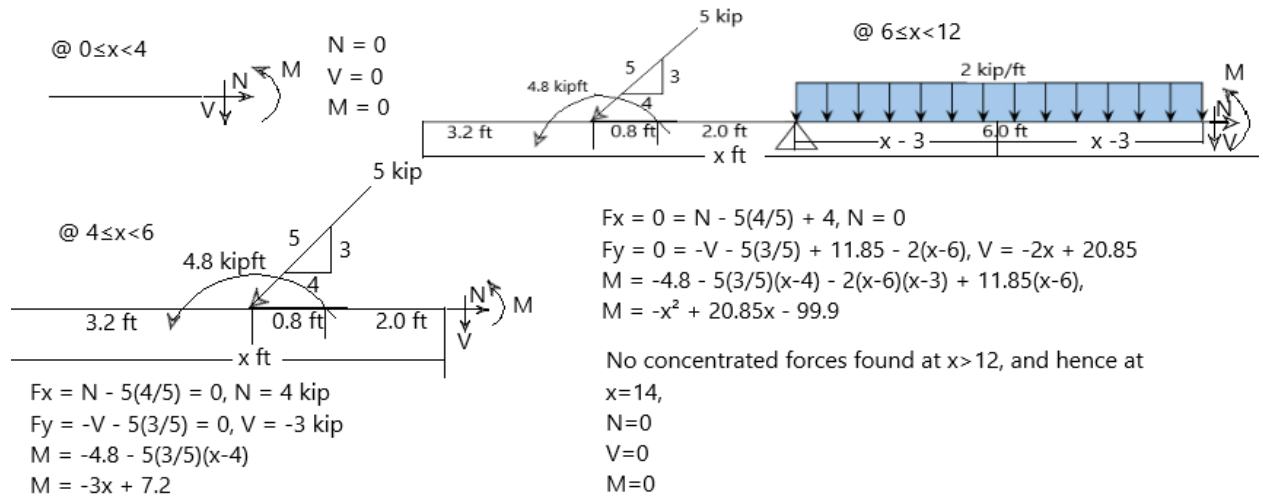
$$b. F_x = N - 5(4/5) = 0, N = 4 \text{ kip}$$

$$c. F_y = -V + 5(3/5) = 0, V = -3 \text{ kip}$$

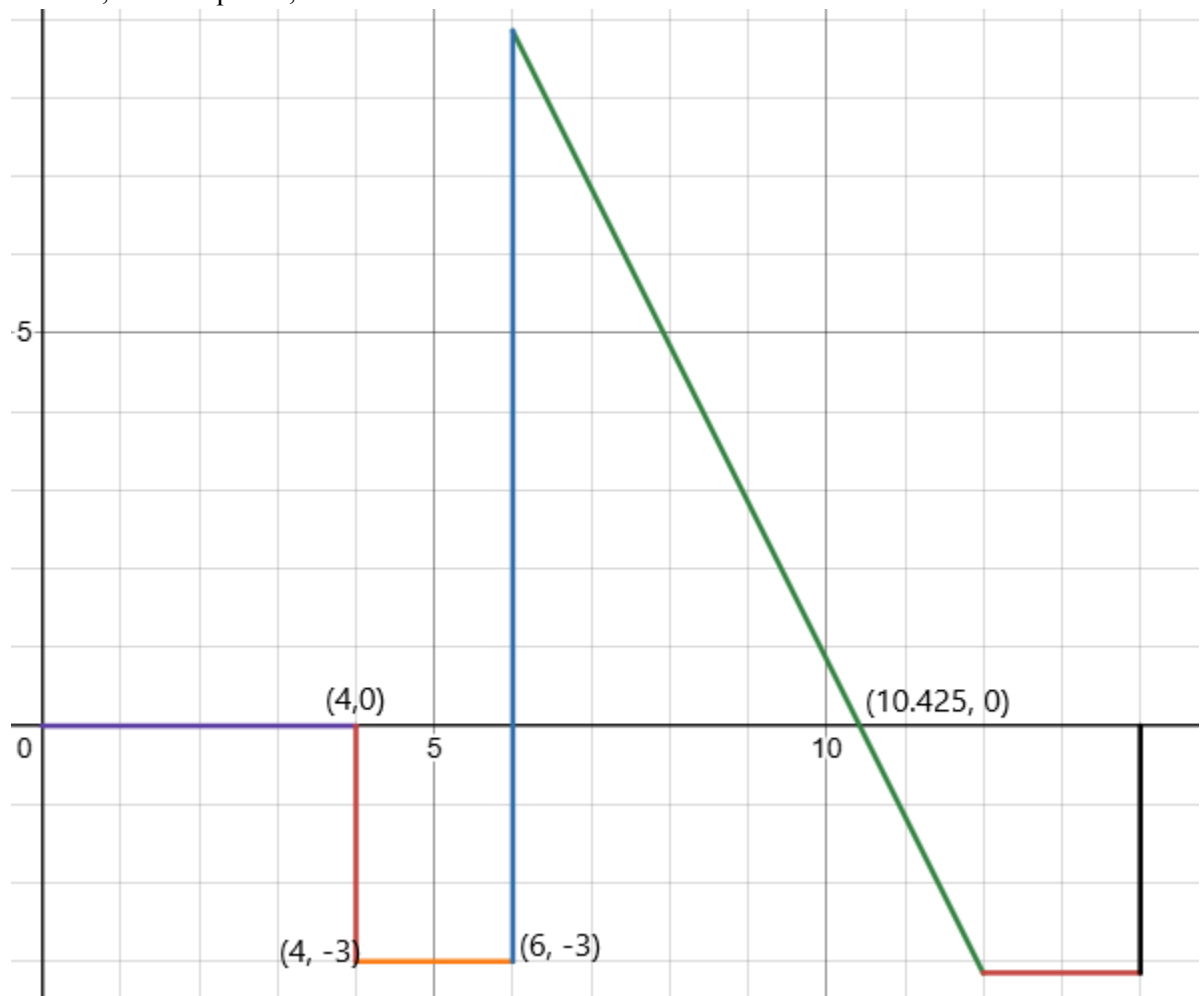
$$d. 4.8 + M = 0, M = -4.8 \text{ kipft}$$

E. (8points) Determine the shear force as a function of x . **Refer to figure below.**

F. (8 points) Determine the bending moment as a function of x .



- G. (6 points) Plot the shear force diagram over the length of the beam, making sure to indicate the extrema, the zero points, and their locations.



- H. (6 points) Plot the bending moment diagram over the length of the beam, making sure to indicate the extrema, the zero points, and their locations

